Binary Detection

June 16, 2026

1 Class Assignments

1.1 Assignment 3: Scalar Detector versus Matched-Filter Detector

Objective. Compare the performance of a single-sample detector with a matched-filter detector for repeated rectangular binary symbols in AWGN.

Problem statement. Assume each transmitted symbol is represented by a rectangular waveform of length L samples:

$$s_k[n] = a_k, \qquad n = 0, 1, \ldots, L-1, \tag{1}$$

where $a_k \in \{a_0, a_1\}$.

The received waveform is

$$r_k[n] = a_k + \eta_k[n], \qquad \eta_k[n] \sim \mathcal{N}(0, \sigma_\eta^2). \tag{2}$$

The scalar detector uses only one sample from each received symbol block:

$$Y_k = r_k[0]. \tag{3}$$

The matched-filter, or integrate-and-dump, statistic is

$$Z_k = \frac{1}{L} \sum_{n=0}^{L-1} r_k[n]. \tag{4}$$

Tasks.

1. Implement the scalar detector using Y_k.
2. Implement the matched-filter detector using Z_k.
3. Repeat the experiment for
$$L \in \{1, 2, 4, 8, 16, 32, 64\}. \tag{5}$$
4. For each value of L, estimate the empirical probability of error for both detectors.
5. Derive the variance of the matched-filter statistic Z_k.
6. Show that averaging over L independent noise samples reduces the effective noise variance to
$$\sigma_Z^2 = \frac{\sigma_\eta^2}{L}. \tag{6}$$
7. Compute the theoretical processing gain:
$$G_{\mathrm{MF}} = 10 \log_{10}(L) \text{ dB}. \tag{7}$$

8. Plot probability of error versus L for both detectors.

Required deliverables.

1. Derivation of the matched-filter statistic.
2. Derivation of the effective noise variance after averaging.
3. Table reporting L, empirical error rate, and theoretical processing gain.
4. Plot of error probability versus L.
5. Brief explanation of why the matched-filter detector outperforms the single-sample detector.